

GEN-AI POWERED DATA ANALYTICS

Internship / Job Simulation Report

Exploratory analytics • Delinquency prediction • Business storytelling • AI-driven collections

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Certificate of Completion

Source document records completion of the Tata GenAI Powered Data Analytics Job Simulation on August 31, 2026.

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Certificate of Completion

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GenAI Powered Data Analytics Job Simulation

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Verification note: the original evidence page instructs the reviewer to verify the user code on the certificate.

PROGRAM

Tata GenAI Powered Data Analytics

ISSUED

Forage

FOCUS

AI-driven tasks

VERIFY

User code on certificate

Simulation Overview

A four-task workflow connecting technical analysis, business communication, and responsible AI design.

ROLE

AI transformation consultant supporting data-driven decision making in financial services.

SCENARIO

Help Geldium Finance address high credit-card delinquency through analytics and AI.

FORMAT

Self-paced Forage job simulation designed to replicate example work tasks performed by Tata iQ professionals.

VALUE

Connect technical analysis with business communication and responsible AI design.

FOUR PRACTICAL TASKS

EDA & risk profiling → delinquency prediction → business report → AI-driven collections

Business Problem & Objectives

Shift from reactive collections to proactive, data-informed risk identification.

THE CHALLENGE

Move from reactive collections toward proactive, data-informed identification of delinquency risk.

THE DECISION

Translate risk signals into clear, measurable and appropriate interventions.

THE CONTROL

Keep fairness, explainability, privacy and human review inside the workflow.

OBJECTIVES

- Understand data quality and customer risk patterns.
- Predict delinquency using a transparent classification approach.
- Communicate insights as a business recommendation.
- Design an adaptive, responsible AI collections workflow.

Task 1 - Exploratory Data Analysis & Risk Profiling

Build the evidence base before predictive modeling.

EDA PURPOSE

Identify patterns, missing values and inconsistencies before predictive modeling.

MISSING DATA

Use justified statistical imputation and validate the choice with domain knowledge.

SYNTHETIC DATA

Can expand datasets while supporting privacy when real customer data is limited or sensitive.

RISK INDICATORS

Recent payment behavior

Missed payments

Credit utilization

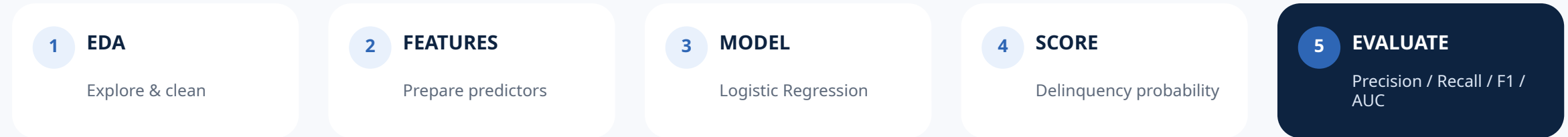
KEY PRACTICE

Treat data quality as part of the modeling problem - not as an afterthought.

Output of this task: a cleaner, better-understood dataset and a defensible risk profile for the next modeling step.

Task 2 - Predicting Delinquency with AI

A transparent classification pipeline focused on probability estimation and balanced evaluation.



TARGET

Binary delinquency outcome. Customer_ID is excluded because it is a unique identifier.

WHY LOGISTIC REGRESSION

Transparent, interpretable and appropriate for a binary probability estimate.

EVALUATION MINDSET

Consider false positives and false negatives, not accuracy alone - especially with imbalanced classes.

DECISION PRINCIPLE

Probability + context + threshold thinking

Evaluation & Responsible Model Use

Model quality is multi-dimensional: classification performance, fairness, explainability and human control.

PRECISION

Among customers predicted delinquent, how many are truly delinquent?

RECALL

How many actual delinquent cases does the model identify?

F1 SCORE

Balances precision and recall.

AUC-ROC

Measures class separability across thresholds.

RESPONSIBLE USE

- Compare outcomes across relevant customer groups for fairness checks.
- Make risk flags understandable to business stakeholders and, where appropriate, affected customers.
- Retain human review for high-impact actions rather than relying on automatic model output.

Task 3 - Business Report & Data Storytelling

Turn analysis into a stakeholder-ready recommendation with a measurable outcome.

GOOD BUSINESS INSIGHT

State the risk pattern → identify the segment → explain the likely impact → connect it to an action.

SMART RECOMMENDATION

Specific • Measurable • Actionable • Relevant • Time-bound

EXAMPLE FROM THE CHECK

6-week SMS outreach pilot for customers under 30 with 2+ missed payments; target a 10% reduction in 30-day delinquency.

BUSINESS COMMUNICATION RULE

Plain language. Clear owner. Measurable outcome.

Designed for a non-technical stakeholder - the recommendation makes the analytical implication operational.

Task 4 - AI-Driven Collections Strategy

Use an adaptive sense-decide-act-learn loop rather than static rules alone.



AUTOMATE Low-impact routine reminders can be automated.

HUMAN-REVIEW High-impact or sensitive interventions should remain human-reviewed.

Core idea: adapt to changing customer context and learn from outcomes instead of following only static rules.

Responsible AI Guardrails

Guardrails are part of the workflow - not a final checklist.

FAIRNESS

Monitor for biased or systematically different outcomes across customer groups.

EXPLAINABILITY

Make model-driven flags and actions understandable to stakeholders.

HUMAN OVERSIGHT

Require review for high-impact decisions and sensitive customer situations.

PRIVACY & COMPLIANCE

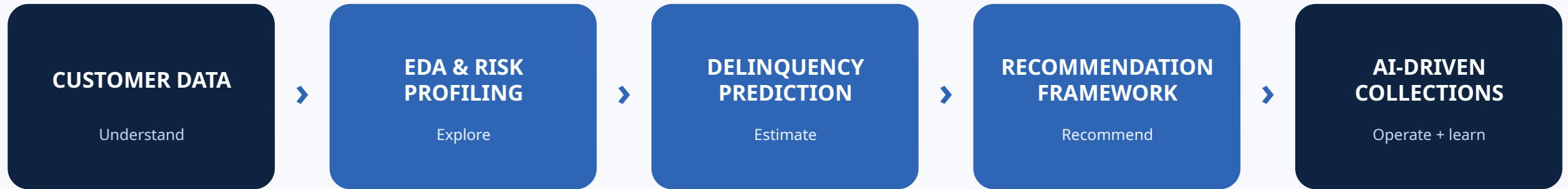
Protect customer information and align with applicable financial and privacy requirements.

CONTINUOUS MONITORING

Track model performance, drift, fairness and customer outcomes; use verified feedback to improve the system.

Integrated End-to-End Workflow

One connected pipeline from customer data exploration to responsible AI-supported collections action.



Pipeline logic

Understand the data → estimate risk → recommend action → operate and learn responsibly

INPUT

Customer-level analytical data and context.

DECISION

Probability plus business rules and thresholds.

FEEDBACK

Outcomes, monitoring and verified feedback inform future actions.

Skills & Professional Capabilities

Technical, business and responsible-AI capabilities developed across the simulation.

DATA ANALYTICS

EDA, data quality, risk profiling

PREDICTIVE ANALYTICS

Model selection, validation, threshold thinking

BUSINESS COMMUNICATION

Data storytelling, SMART recommendations

AI STRATEGY

Agentic AI, automation, learning loops

RESPONSIBLE AI

Fairness, explainability, privacy, human oversight

GENAI

Using GenAI to structure analysis and business outputs

Key Learning Outcomes

The simulation reinforced that strong analytics is a workflow - not a single model.

- 1 Connected EDA, predictive modeling and business recommendations into one analytics workflow.

- 2 Learned that data quality and missing-value handling directly influence model reliability.

- 3 Practised translating model outputs into plain-language actions for business stakeholders.

- 4 Learned how agentic AI can support collections while keeping humans accountable for high-impact decisions.

- 5 Developed a stronger understanding of fairness, explainability, privacy and responsible AI in financial use cases.

Conclusion & Future Scope

A useful AI solution combines reliable data, understandable insights, measurable actions, responsible automation and a feedback loop.

KEY TAKEAWAY

AI value comes from the full operating system around the model.

Reliable data + understandable insights + measurable action + responsible automation + feedback

FUTURE SCOPE

- Experiment with additional classification models.
- Calibrate thresholds and test intervention strategies.
- Monitor fairness and model / data drift in production-style settings.
- Use controlled intervention tests to validate outcomes.

NEXT-LEVEL VIEW

From a job simulation workflow toward production-grade model governance.

What the Simulation Produced

A compact view of the work products implied by the four practical tasks.

01

EDA & RISK PROFILE

Data quality checks
Risk indicators
Missing-value handling
Model-ready predictors

02

DELINQUENCY MODEL

Binary target
Logistic Regression
Probability score
Precision / Recall / F1 / AUC

03

BUSINESS REPORT

Risk pattern
Segment identification
Likely impact
SMART recommendation

04

AI COLLECTIONS

Sense context
Combine model + rules
Trigger outreach
Learn + adapt

Reference basis All content in this deck is re-presented from the supplied 15-page reference PDF, with the student name and roll number updated as requested.

Thank You

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Tata Group x Forage | GenAI Powered Data Analytics

B.Tech (CSE), 5TH SEMESTER

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